Pretraining, Fine-Tuning, and Instruction Tuning

How language models acquire and adapt capabilities

Purpose

This short reference note is written for learning, semantic retrieval, and grounded question-answering experiments. It uses original explanatory content and is organized into coherent sections rather than unrelated filler.

1. Pretraining

Pretraining exposes a model to a large collection of text or other data and optimizes it for a learning objective. Many autoregressive language models learn to predict the next token given preceding tokens. Through this objective, the model learns statistical patterns in language, facts represented in its data, and structures useful for generating text. Pretraining is computationally expensive and the resulting model can still produce false statements or reflect limitations in its training data.

2. Fine-Tuning

Fine-tuning continues training from an existing model using a more targeted dataset. The goal might be to improve performance in a domain, follow a task format, or adapt to examples. Full fine-tuning updates many or all model parameters; parameter-efficient approaches update a smaller set of parameters or adapter modules. Fine-tuning requires high-quality examples and evaluation. Poor data can cause overfitting, narrow behavior, or regression on previously learned capabilities.

3. Instruction Tuning

Instruction tuning uses examples that pair instructions with desired responses. It aims to make a model more helpful at following user requests and producing expected formats. It is a form of supervised fine-tuning, though deployed systems may involve additional preference optimization or safety training. Instruction tuning does not guarantee that the model will obey every instruction or always distinguish evidence from speculation.

4. Fine-Tuning Versus RAG

Fine-tuning changes model parameters; retrieval-augmented generation (RAG) supplies relevant external passages at answer time. Fine-tuning can shape behavior, style, or task performance. RAG is often more convenient when facts change frequently or answers must cite a private document collection. The approaches can be combined, but they solve different problems. If a system must answer questions about newly uploaded PDFs, retrieval is usually the first mechanism to evaluate before considering fine-tuning.

5. Data Quality and Evaluation

Training examples should be relevant, consistent, legally usable, and representative of real use. Separate training data from evaluation data to avoid measuring memorization. Evaluate not only average accuracy but also edge cases, unsupported questions, format compliance, and regressions. Keep a baseline model and configuration so changes can be compared fairly.

6. Choosing a Strategy

Use prompting when a task can be handled through clear instructions and examples. Use RAG when the answer depends on a searchable knowledge base or source traceability. Consider fine-tuning when there is a repeated behavior or domain pattern that prompting and retrieval do not adequately address, and when a suitable training set and evaluation budget exist. Decisions should be based on measured failure modes, not on the assumption that more training always improves results.

Key Takeaways

Pretraining builds broad language capabilities, fine-tuning adapts behavior, and instruction tuning emphasizes following instructions. RAG supplies external evidence at inference time and is particularly useful for document-grounded question answering.

End of document • Use source passages as evidence and verify claims during evaluation.